

Financial Functions,

1. FV - future value.

FV calculates how much an investment or loan will be worth in the future after a certain period.

syntax : $=FV(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}])$

rate = interest rate per period

nper = Number of periods

pmt = payment made each period

[pv] = present value / money which is invested today

2. PMT - payment / EMI.

PMT calculates the regular payment / EMI required for a loan or investment

syntax : $=PMT(\text{rate}, \text{nper}, \text{pv}, [\text{fv}])$

3. PPMT : Principal payment

calculates a principal portion of a particular loan payment

When we pay EMI, the payment contains:

* principal amt = The original amt of money borrowed or invested, before adding interest.

→ Interest amt = The extra money paid or earned on the principal amount.

Ex: we have a loan of £50,000
monthly EMI is £2000

Suppose for the first month

→ $PPMT = £1,500$ → reduces our loan from £50,000 to £48,500

→ $IPMT = £500$ → goes towards interest.

Syntax : $= PPMT (rate, per, nper, pv)$

per = it tells you which specific payment / period you want to calculate.

Ex: • $per = 1$ → 1st month
• $per = 5$ → 5th month

$Nper$ = How many period in total, tells you the total number of payments / periods.

Ex: • 3 year loan with monthly payments.
• $NPER = 3 \times 12 = 36$ months

per → which one?

$nper$ → How much all together?

4. IPMT - Interest payment.

calculates the interest portion of a particular loan payment

Syntax: =IPMT (rate, per, nper, pv)

5. RATE - Interest rate

calculates the interest rate per period for a loan or investment

Syntax: =RATE (nper, pmt, pv, [fv])

6. PV - Present value.

calculate the current value of the money based on a future amt, interest rate and time.

Syntax: =PV (rate, nper, pmt, [fv])

7. NPER - Number of periods.

calculates how long it will take to reach a particular financial goal or repay a loan

Syntax: =NPER (rate, pmt, pv, [fv])

period: one payment / investment interval, such as month or year

RD - Recurring Deposit a fixed amount deposited regularly, usually every month, to earn interest.

Questions.

1. Mr Sunil purchased a hero honda Karizma for Rs. 65000 by taking a loan from ICICI Bank, repayable in 3 years by monthly installments with interest @ 8.5% p.a. Calculate the EMI for the loan [PMT]

$$= \text{PMT} (8.50/12, 3*12, 65000)$$

$$= \text{PMT} (\text{rate}, \text{nper}, \text{pv}) \quad \text{o/p : 2051}$$

2. Mr Sunil wants to know the breakup of the principal amt in the EMI payable for the 1st month [PPMT]

$$= \text{PPMT} (\text{rate}, \text{per}, \text{nper}, \text{pv}, [\text{fv}])$$

$$= \text{PPMT} (8.50/12, 1, 3*12, 65000) \quad \text{o/p: 1591}$$

3. Now, Mr. Sunil wants to know the breakup of the interest amt in the EMI payable for the 1st month [IPMT]

$$= \text{IPMT}(\text{rate}, \text{per}, \text{nper}, \text{pv}, [\text{fv}])$$

$$= \text{IPMT}(8.50/12, 1, 3 \times 12, 65,000) \quad \text{OIP: 460}$$

4. Ms. Meenakshi has a fixed deposit of Rs. 25000 in HDFC bank for 5 years on which interest given by the bank is 8.25%. Calculate the maturity value of Meenakshi [FV]

$$= \text{FV}(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}])$$

$$= \text{FV}(8.25/12, 5, 25000) \quad \text{OIP: 37,160}$$

5. Ms. Rita is having the option of going for RD in which amount payable per month is Rs. 1200 interest @ 7.5% and the period of deposit will be 5 years, she asked you to calculate the maturity value for this option [FV]

$$= \text{FV}(\text{rate}, \text{nper}, \text{pmt}, [\text{pv}])$$

$$= \text{FV}(7.5/12, 5 \times 12, 1200) \quad \text{OIP: 87,032}$$

6. Mr. JP wants to invest in a plan where he will pay Rs. 1476 quarterly for 10 years to get a maturity value of Rs. 80,000. He wants you to calculate the rate of interest he will earn on this plan of investment. [RATE]

$$= \text{RATE}(\text{nper}, \text{pmt}, \text{pv}, \{\text{fv}\})$$

$$= \text{RATE}(10 \times 4, , 1476, 80000)$$

o/p: 10.5%

7. Before opting for the plan with the rate calculated above and for the same period of maturity value. Mr. JP now wants to know the present value of the investment plan. [PV]

$$= \text{PV}(\text{rate}, \text{nper}, \text{pmt}, \{\text{fv}\})$$

$$= \text{PV}(10.50\%, 10 \times 4, , 80000)$$

o/p: -1476

8. Mrs. Bini wants to go for a RD scheme in any bank with an amt of ₹1520 per month where the interest is payable @ 7.5% per annum, she needs your help to find the no of installments she has to pay (also calculate the no of years) to get a maturity value of ₹ 50000 [NPER]

$$= \text{NPER}(\text{rate}, \text{pmt}, \text{pv}, \{\text{fv}\})$$

$$= \text{NPER}(7.5/12, 1520, , 50000)$$

o/p: 30